

USE-CASE FLOW

UC-02: Generate Weekly Meal Plan

Primary Actor: Subscriber **Supporting Actor:** Dietitian

Preconditions

- The subscriber is authenticated.
- The subscriber has a saved dietary profile containing allergies, diabetic limits, calorie goals, and macro-nutrient constraints.
- The subscription is active for the requested week.

Postconditions

- A weekly meal plan is generated according to the subscriber's constraints.
- The plan is saved for dietitian review.
- No excluded allergen is included in the generated menu.

Main Success Scenario

1. Subscriber selects "Generate Weekly Meal Plan."
2. System retrieves the subscriber's dietary profile and active subscription details.
3. System validates the stored dietary constraints and allergen exclusions.
4. System generates a weekly menu satisfying the configured calorie and macro-nutrient limits.
5. System checks the generated menu for excluded allergens and constraint violations.
6. System saves the generated meal plan with status "Pending Dietitian Review."
7. System notifies the dietitian that a meal plan is ready for review.
8. Dietitian reviews the meal plan and approves it.
9. System changes the plan status to "Approved" and makes it available for the scheduled subscription.

Alternate Flow - Constraint Violation

- System rejects the generated menu.
- System identifies the violated constraint or excluded ingredient.
- System regenerates the menu using valid constraints.
- If a compliant plan cannot be generated, the system displays an error and flags the case for dietitian intervention.